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Unit 1 : Session 8 : SLO 1 | Asymptotic Notations Big O, Big Omega, Big Theta

I. Fill in the Blanks

1. Big O is the upper bound on running time.
2. Big Omega (Ω) is the lower bound.
3. Big Theta (Θ) is the tight bound.
4. worst Big O analyzes worst-case performance.
5. best Big Omega analyzes best-case performance.
6. upper and lower Θ bounds from both sides.
7. $O(n)$ is linear search.
8. $O(\log n)$ is binary search.
9. $O(n^2)$ is bubble sort worst case.
10. O if $f(n) = 3n + 2$ is $O(n)$.
11. O if $f(n) = 5n^2 + 3n + 7$ is $O(n^2)$.
12. lower-order constants and lower-order terms are ignored.
13. $O(1)$ (constant) is same time regardless of input size.
14. Θ if $f(n) \approx c \cdot g(n)$ for $n \approx n_0$ means $f(n) = \Theta(g(n))$.
15. O and Θ if $\Theta(g(n))$ means both $O(g(n))$ and $\Omega(g(n))$.